Closed sets and limit points

Defn: Closed set.

A closed set, C subset of a topological space $(X, \mathcal{T})$ is called closed if $X \setminus C$ is open.

Note that a set can be both open and closed.

Theorem: Properties of Closed Sets:

Let $X, \mathcal{T}$ be a topological space. Then

1) $X, \emptyset$ are closed.
2) $\bigcap_{\alpha \in A} C_\alpha$ is closed if C_α is closed
3) $\bigcup_{i=1}^{n} C_i$ is closed if each C_i is closed

Proof: Let X be a topological space, C_α closed.

$X \setminus \bigcap_{\alpha \in A} C_\alpha = \bigcup_{\alpha \in A} (X \setminus C_\alpha)$ which is an arbitrary union of open sets $\Rightarrow$ open. So, $\bigcap_{\alpha \in A} C_\alpha$ is closed. $X \setminus \bigcup_{\alpha=1}^{n} C_\alpha = \bigcap_{\alpha=1}^{n} (X \setminus C_\alpha)$ which is a finite intersection of open sets, so open. Then $\bigcup_{\alpha=1}^{n} C_\alpha$ is closed. $\emptyset$ is open, so X is closed and X is open so $\emptyset$ is closed

Monday, September 14, 2026 5:33 PM

□

If we have $Y \subseteq X$ with the subspace topology, $\mathcal{T}_Y$, C is closed in Y means C is closed in $\mathcal{T}_Y$.

Theorem: Let $Y \subseteq X$ be a subspace of X. Then C is closed in Y iff $C = D \cap Y$ for a closed set D in X.

This shows that if C closed in Y and Y closed in

X, Then C closed in X.

Defn: Let A be a subset of a topological space X. Then:

1) The interior of A, $\text{int}A$ is the union of all open subsets of A.
2) The closure of A, clA or $\bar{A}$, is the intersection of all closed sets containing A.

Note that $\text{int}A \subseteq A \subseteq \bar{A}$.

Theorem: Properties of interior and Closure.

Let $A \subseteq X$, (X, τ)

1) $\text{int}A$ is the largest open set contained in A.
2) clA is the smallest closed set containing A.
3) A open $\iff A = \text{int}A$.
4) A closed $\iff A = \bar{A}$.
5) If $Y \subseteq X$ and $A \subseteq Y$, then the closure of A in Y is given by $\bar{A} \cap Y$.

Terminology:

- A intersects $B \iff A \cap B \neq \emptyset$.
- U is a neighborhood of $x \iff U$ open and $x \in U$.

Theorem: Closure via neighborhood.

Let A be a subset of a topological space X.

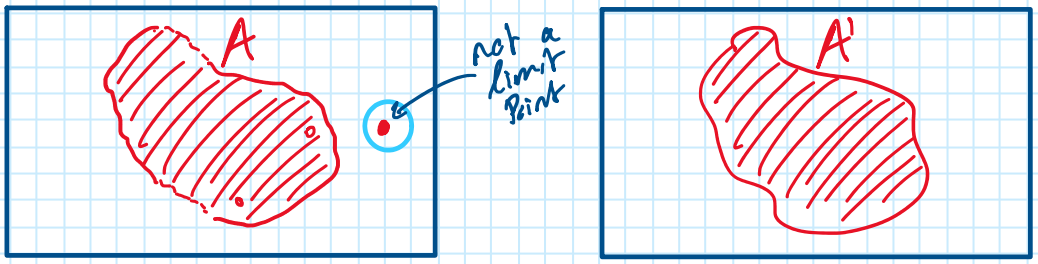
a) $x \in \bar{A} \iff$ Every neighborhood of x intersects A.

b) Suppose that $\mathcal{B}$ is a basis for τ. Then: $x \in \bar{A} \iff$ Every $B \in \mathcal{B}$ with $x \in B$ intersects A.

Defn: limit point.

If A is a subset of a topological space X

If A is a subset of a topological space X and $x \in X$, then x is a limit point of A if for every neighborhood U of x we have $U \setminus \{x\} \cap A \neq \emptyset$. The set of limit points of A is denoted A'.



Notice that isolated points are not limit points. x need not be in A.

$x \in A' \iff x \in \overline{A \setminus \{x\}}$.

Theorem: Closure via limit points.

Let A be a subset of a topological space X. Then:

1) $\bar{A} = A \cup A'$.

2) A closed $\iff$ A contains its limit points $\iff A' \subseteq A$.

Proof: We first show that $\bar{A} \supseteq A \cup A'$. Clearly, $A \subseteq \bar{A}$, so $A \cup A' \subseteq \bar{A}$. Now suppose $x \in A'$. Then for every neighborhood U of x, we have $U \setminus \{x\} \cap A \neq \emptyset \Rightarrow U \cap A \neq \emptyset \Rightarrow x \in \bar{A}$.

Now let $x \in \bar{A}$. If $x \in A$, then obviously $x \in A \cup A'$. Assume then that $x \in \bar{A} \setminus A$. Therefore every neighborhood U of x intersects A, but not in x since $x \notin A$. Thus, $A \cap U \setminus \{x\} \neq \emptyset$. Thus, $x \in A' \subseteq A \cup A'$. $\therefore \bar{A} = A \cup A'$.

2) A is closed $\Rightarrow A = \bar{A}$. From (1), $\bar{A} = \bar{A} \cup A'$. Thus, $A' \subseteq \bar{A} = A$. □

Exercises:

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2) Show that A closed in Y and Y closed in X implies A closed in X.

Proof: Suppose A closed in Y.
So $A = C \cap Y$, for some closed set C in X.
C and Y are closed in X, so $C \cap Y = A$ is closed in X.

□

3) Show that if A closed in X, B closed in Y, then $A \times B$ is closed in $X \times Y$.

Proof: Suppose A closed in X, B closed in Y.

$(X \times Y) \setminus (A \times B) = (X \setminus A \times Y) \cup (X \times Y \setminus B)$

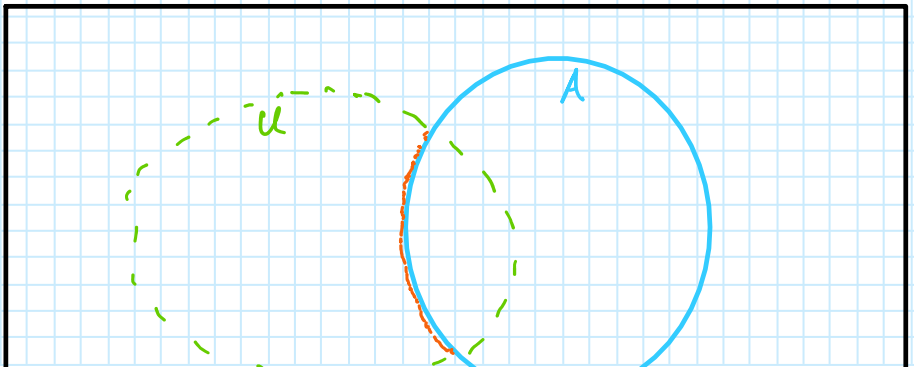
We must show these are both open in $(X \times Y)$.
For $P \in X \setminus A \times Y$, $\exists B \in \mathcal{B}_x$ with $\forall x \in X \setminus A$, $x \in B \subseteq X \setminus A$. Y is a basis element for Y by defn.
So, for any $P \in X \setminus A \times Y$, $P_1 = x$, $P_2 = y \in Y$, then

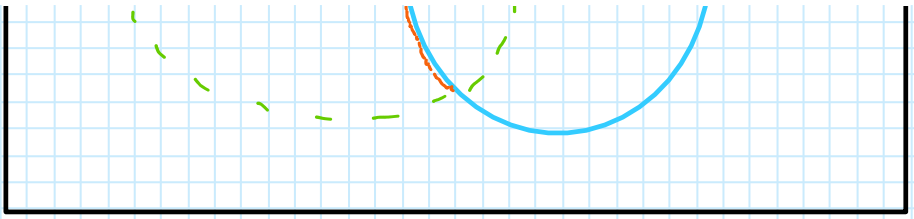
$P \subseteq B \times Y \subseteq X \setminus A \times Y$. Thus, $X \setminus A \times Y$ is open in the product topology. The case for $(X \setminus Y \setminus B)$ is analogous. Thus, $(X \times Y) \setminus (A \times B)$ is open $\Rightarrow$ $A \times B$ is closed.

□

4) Show that if U is open in X and A is closed in X, then $U \setminus A$ is open in X and $A \setminus U$ is closed in X.

Proof: Suppose U open, A closed.





Notice that $(X \setminus A) \cap U$ is the same as $U \setminus A$. $X \setminus A$ is open and U is open, so $(X \setminus A) \cap U = U \setminus A$ is open.

Similarly $(X \setminus U) \cap A = A \setminus U$. $X \setminus U$ is closed, A is closed, so $(X \setminus U) \cap A = A \setminus U$ is closed.

□

5) Let X be an ordered set in the order topology. Show that $\overline{(a,b)} \subseteq [a,b]$.

Proof: $[a,b]$ is the set of x s.t. $a \leq x \leq b$, and (a,b) is the set of x s.t. $a < x < b$.

Suppose $x \in \overline{(a,b)}$. Then for all neighborhoods U of x, $U \cap (a,b) \neq \emptyset$. In the order topology, for all $\varepsilon > 0$, $(x-\varepsilon, x+\varepsilon) \cap (a,b) \neq \emptyset$. Suppose $x \in (a,b)$. Then clearly $x \in [a,b]$. Suppose then that $x \notin (a,b)$, $x \in \overline{(a,b)}$. Then $x > b$ or $x < a$. Suppose without loss that $x < a$. By definition of closure, $(x-\varepsilon, x+\varepsilon) \cap (a,b) \neq \emptyset$. So, $x + \varepsilon > a$. $\Rightarrow x > a - \varepsilon$. But $a > x$. So $a - \varepsilon < x < a$ for all ε means $x = a$. So $x \in \{a\} \cup (a,b)$, which is $x \in [a,b)$. The other side is identical so. $x \in \overline{(a,b)} \Rightarrow x \in [a,b]$.

□

6) Let A, B, A_α be subsets of X. Prove that:

a) $\overline{A \cup B} = \bar{A} \cup \bar{B}$.

b) $\overline{\bigcup A_\alpha} \supseteq \bigcup \bar{A}_\alpha$; give example where equality fails

Proof: a) Suppose $x \in \overline{A \cup B}$. Then for all U neighborhood of

Proof: a) Suppose $x \in \overline{A \cup B}$. Then for all U neighborhood of x, $U \cap (A \cup B) \neq \emptyset$. $U \cap (A \cup B) = (U \cap A) \cup (U \cap B) \neq \emptyset$. So, x must be in the closure of A or B. So, $x \in \bar{A} \cup \bar{B}$. Now show that $\overline{A \cup B} \subseteq \bar{A} \cup \bar{B}$. Now suppose $x \in \bar{A} \cup \bar{B}$. Then x must be in the closure of A or the closure of B. Suppose $x \in \bar{A}$. Then $\forall U$ of x, $U \cap A \neq \emptyset$. So, $(U \cap A) \cup (U \cap B) = U \cap (A \cup B) \neq \emptyset$. So, $x \in \overline{A \cup B}$. The case where $x \in \bar{B}$ is identical, so $\overline{A \cup B} = \bar{A} \cup \bar{B}$.

b) Let $x \in \bigcup \bar{A}_\alpha$. Then for some A_j, $\forall U$ of x, $U \cap A_j \neq \emptyset$. Clearly $A_j \subseteq \bigcup A_\alpha$, so $U \cap A_j \subseteq U \cap (\bigcup A_\alpha)$. Because $U \cap A_j$ is nonempty and contained in $U \cap (\bigcup A_\alpha)$, $\bigcup A_\alpha \neq \emptyset$. So $x \in \overline{\bigcup A_\alpha}$. Thus, $\bigcup \bar{A}_\alpha \subseteq \overline{\bigcup A_\alpha}$.

Consider an increasing sequence of sets, $A_\alpha = \left(\frac{1}{\alpha}, 1 - \frac{1}{\alpha}\right)$. Clearly in the order topology, $\{0\}, \{1\} \in \overline{\bigcup A_\alpha} = (0,1)$. But $\forall \alpha > 0$, $\{0\}$ and $\{1\}$ are not in the closure of $\left(\frac{1}{\alpha}, 1 - \frac{1}{\alpha}\right)$.

7) It is not true that U must intersect some A_α in our infinite union.

8)

a) Let $x \in \overline{A \cap B}$. Then $\forall U$ of x, $U \cap A \cap B \neq \emptyset$. $(U \cap A) \cap (U \cap B) = U \cap A \cap B$. So $U \cap A \neq \emptyset$ and $U \cap B \neq \emptyset$. So $x \in \bar{A}$ and $x \in \bar{B}$. So $\overline{A \cap B} \subseteq \bar{A} \cap \bar{B}$.

Suppose that $x \in \bar{A} \cap \bar{B}$. Then $\forall U$ of x, $U \cap A$ and $U \cap B$. So $U \cap A \cap U \cap B = U \cap A \cap B \neq \emptyset$. So $\bar{A} \cap \bar{B} \subseteq \overline{A \cap B}$.

Equality is true.

b) Later...

c) Let $A = [0,1]$, $B = (0,1)$. Then $\overline{A - B} = \{0\} \cup \{1\}$. But $\bar{A} - \bar{B} = \emptyset$ (in the order topology at least), so

c) Let $A = [0,1]$, $B = (0,1)$. Then $A - B = \{0\} \cup \{1\}$. But $\overline{A} - \overline{B} = \emptyset$ (in the usual topology at least), so $\overline{A} - \overline{B} \not\supseteq \overline{A - B}$. For $x \in \overline{A} - \overline{B}$. Then $\forall U_x$, $U_x \cap A \setminus U_x \cap B \neq \emptyset$. $(U_x \cap A) \setminus (U_x \cap B) \neq \emptyset$. So, $B \not\supseteq A$. So there must be $y \in A$ s.t. $y \in U_x \cap A$ and $y \notin U_x \cap B$. y must be in U_x, therefore it must not be in B. So, $y \in U_x \cap (A \setminus B)$. So, $U_x \cap (A \setminus B) \neq \emptyset$. Then, $\overline{A} - \overline{B} \subseteq \overline{A - B}$.

d) $(A \cup B)' = A' \cup B'$?

Suppose $x \in (A \cup B)'$. Then $\forall U_x$, $U_x \cap [(A \cup B) \setminus \{x\}] \neq \emptyset$. $(A \cup B) \setminus \{x\} = (A \setminus \{x\}) \cup (B \setminus \{x\})$. So, $U_x \cap [(A \setminus \{x\}) \cup (B \setminus \{x\})]$. and furthermore $[U_x \cap (A \setminus \{x\})] \cup [U_x \cap (B \setminus \{x\})] \neq \emptyset$. Then, $x \in A' \cup B'$. So $(A \cup B)' \subseteq A' \cup B'$. Suppose $x \in A' \cup B'$. Then $\forall U_x$, $U_x \cap [(A \setminus \{x\}) \cup (B \setminus \{x\})] \neq \emptyset$. is equivalent to $U_x \cap [(A \cup B) \setminus \{x\}] \neq \emptyset$. So $A' \cup B' \subseteq (A \cup B)'$.

e) $(A \cap B)' = A' \cap B'$?

$(A \cap B)' \not\supseteq A' \cap B'$. Consider $(-\infty, a) = A$, $(a, \infty) = B$. Then $(A \cap B)' = \emptyset' = \emptyset$. But $A' = \{a\}$, $B' = \{a\}$. So $A' \cap B' = \{a\}$. Suppose $x \in A' \cap B'$. Then $x \in A'$ and $x \in B'$. So $\forall U_x$, $U_x \cap A \setminus \{x\} \neq \emptyset$ and $U_x \cap B \setminus \{x\} \neq \emptyset$. So $A \setminus \{x\}$ and $B \setminus \{x\}$ have elements of U_x. — contradiction perhaps.

Suppose $\exists x \in (A \cap B)'$, but $x \notin A' \cap B'$. Then $\forall U_x$, $U_x \cap (A \cap B) \setminus \{x\} \neq \emptyset$. But $\exists U_x^*$ with $U_x^* \cap (A \setminus \{x\}) \cup U_x^* \cap (B \setminus \{x\}) = \emptyset$. So $U_x^* \cap [(A \cup B) \setminus \{x\}] = \emptyset$. Clearly $(A \cup B) \setminus \{x\} \supseteq (A \cap B) \setminus \{x\}$. So $U_x^* \cap [(A \cup B) \setminus \{x\}] = \emptyset$ contains $U_x^* \cap [(A \cap B) \setminus \{x\}]$. This is a contradiction. So, $A' \cap B' \supseteq (A \cap B)'$.

□

g) Let $A \subseteq X$, $B \subseteq Y$. Show that in $X \times Y$, $\overline{A \times B} = \overline{A} \times \overline{B}$.

9) Let $A \subseteq X$, $B \subseteq Y$. Show that in $X \times Y$, $\overline{A \times B} = \bar{A} \times \bar{B}$.

Proof: Suppose $P \in \overline{A \times B}$. Then $\forall U_p$, $U_p \cap A \times B \neq \emptyset$. $P = (x, y)$, where $x \in X$ and $y \in Y$. U_p intersects $A \times B$ implies that $\pi_1(U_p) \cap A$ and $\pi_2(U_p) \cap B$ are both nonempty. $\pi_1(U_p)$ is a neighborhood of x, and $\pi_2(U_p)$ is a neighborhood of y. So x is in the closure of A, and y is in $\bar{B}$. Thus $x \times y = P \in \bar{A} \times \bar{B}$. So, $\bar{A} \times \bar{B} \subseteq \overline{A \times B}$.

Suppose $P \in \bar{A} \times \bar{B}$. Then $P_1 = x \in \bar{A}$, $P_2 = y \in \bar{B}$. Let U_p be a neighborhood of P. We must show that $U_p \cap A \times B \neq \emptyset$. Let $U_x \subseteq X$ and $U_y \subseteq Y$ be neighborhoods of x and y. Then $U_x \times U_y$ is a neighborhood of P. Furthermore, for any U_p, we can pick U_x and U_y s.t. $U_p \supseteq U_x \times U_y$. So, it suffices to show that $U_x \times U_y \cap A \times B$ is nonempty. Because x, y are in $\bar{A}, \bar{B}$, $U_x \cap A \neq \emptyset$ and $U_y \cap B \neq \emptyset$. So $U_x \times U_y \cap A \times B \neq \emptyset$. Thus $U_p \cap A \times B \neq \emptyset$. So, $P \in \overline{A \times B}$, $\bar{A} \times \bar{B} \subseteq \overline{A \times B}$.

Thus, $\overline{A \times B} = \bar{A} \times \bar{B}$.

□